Exposure-Controlled Data Minimization for Personalized LLM Prompts

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ABSTRACT

Personalized LLM assistants expose personal history when retrieved records enter a provider's prompt. We calibrate record admission with conformal risk control, bounding auditor-relative expected *profile exposure* per prompt by α for users exchangeable with calibration users. Any ranking can be used inside the monotone admission rule. Counting the excluded query as one record gives a worst-case total-prompt bound of $\alpha + 1$. On LaMP-QA, LaMP-3 and LaMP-7, calibrating on one LLM auditor can exceed the target under another auditor's labels; union calibration controls both by construction. On LaMP-3, where retrieval helps, a strict exposure budget costs utility across three readers. At $\alpha = 0.5$, cleared similarity keeps 99% [83, 114] of a 14B reader's gain with less than half the flagged records of top-5 retrieval. Removing flagged text while preserving ratings retains most of that gain. These are model-audited, single-query results; human validation and real multi-query disclosure measurement remain open.

CCS CONCEPTS

• **Security and privacy** → **Privacy protections**; • **Information systems** → **Personalization**.

KEYWORDS

data minimization, personalization, conformal risk control, retrieval-augmented generation, privacy

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1 INTRODUCTION

Personalized assistants answer better when the prompt carries records from the user's history such as earlier posts, reviews or messages [24, 25]. When the model runs at a provider, each admitted record leaves the user's device, and personal histories are full of self-disclosures about health, family, work and location [8]. Data minimization asks that personal data be limited to what the purpose needs [5, 11], which here means an on-device gate between the profile and the prompt.

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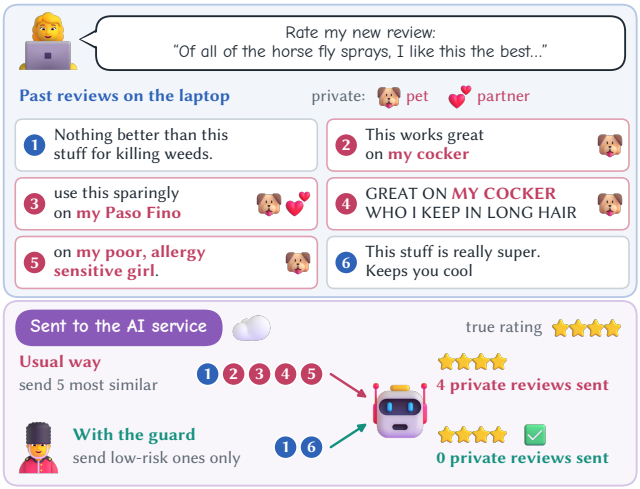


Figure 1: The usual prompt sends four private reviews of this LaMP-3 user. The guard sends only low-risk ones, with a cut-off tuned on 225 other users to pass at most half a private review on average, and the answer is still 4 stars.

Existing gates select profiles or personal fields for a task, reporting benchmark utility and leakage with hand-set or utility-tuned thresholds [3, 9, 16, 29, 30, 32]. Deployment instead needs an exposure bound at a chosen level for unseen users.

We obtain that number with conformal risk control [2]. The exposure of an admission rule, the count of admitted records that an auditor flags as self-disclosing, is bounded by the record cap and monotone in the rule's one parameter, so calibrating that parameter on audited users bounds the expected exposure of a new user's prompt by α for any ranking of records (Figure 1). The statement is per prompt, in expectation over users exchangeable with the calibration sample, and relative to the auditor, which runs offline on consenting calibration users only. The theory is standard; we study its use for record admission and its utility cost.

Across three tasks, two LLM auditors and three readers, we compare calibrated admission with retrieval and a hindsight oracle. The experiments test the bound's dependence on auditors and population, and measure the utility cost of restricting whole records rather than removing disclosures within them.

2 RISK-CONTROLLED RECORD ADMISSION

Threat model. A user's profile P of personal records stays on the device, together with a retriever, the admission rule and its scores. For a query q the rule places a set $A(q, P)$ of at most K records from a candidate pool $C(q, P)$ in the prompt of a reader LLM run by a remote provider. Whatever is admitted leaves the device, and the

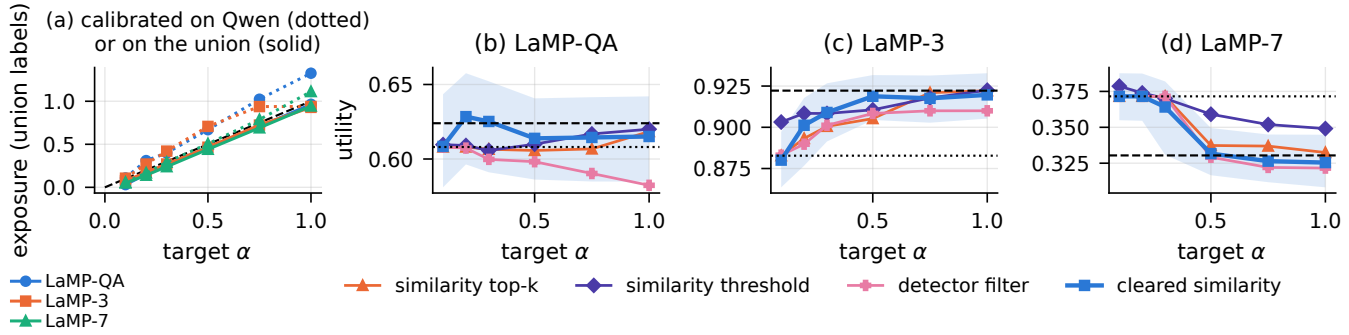


Figure 2: (a) Exposure under union labels of test users, worst of four rules, for rules calibrated on the primary auditor (dotted) or on the union (solid). (b–d) Cross-fitted utility of the rules calibrated on union labels, with a 95% band for cleared similarity. Dotted is no profile, dashed top-5.

provider is assumed to see the full prompt. The utility $U(q, S)$ of a set S is the task metric of the reader's output.

Exposure. An *auditor* sets $a(r) = 1$ when record r contains a self-disclosure, a statement revealing personal information about the author or a person close to them [8]. We control two losses of an admitted set, the count $E(S) = \sum_{r \in S} a(r) \in [0, K]$ and the indicator $Z(S) = \mathbb{1}[E(S) \geq 1]$ that the prompt carries any disclosure. The auditor may be a large model or a human annotator. It labels a sample of consenting calibration users offline and never runs on the device. This is incremental profile exposure, excluding the query sent by every rule, even the empty rule. Counting a query as one additional record gives $E_{\text{total}} = E(S) + a(q)$ and only $\mathbb{E}E_{\text{total}} \leq \alpha + 1$ without a query audit. A total-prompt budget needs query labels and an abstention or local-only fallback.

Admission families. We calibrate any family $\{A_\lambda\}$ with an ordered parameter on a finite grid $\lambda_0 < \cdots < \lambda_J$ and $A_{\lambda_0} = \emptyset$. A *top-k* rule admits the k highest-scoring candidates. A *threshold* rule admits up to K candidates whose score is at least a cut-off that decreases along the grid, so the number of records adapts to the query. A *cleared* rule uses an on-device sensitivity proxy $s(r) \in [0, 1]$ and admits

$$A_\lambda(q, P) = \text{top-}K_g \{r \in C(q, P) : s(r) \leq \lambda\}, \tag{1}$$

where g ranks candidates, by similarity to q in our main rule. Top-k and threshold rules are nested. In a cleared rule a newly cleared record can displace a less highly ranked one, so we control the running maximum $L(\lambda_j) = \max_{j' \leq j} \ell(A_{\lambda_{j'}})$ of the loss $\ell \in \{E, Z\}$, which is nondecreasing, bounded by $B = K$ for E and $B = 1$ for Z, and never below ℓ.

Calibration. Let $L_1, \ldots, L_n$ be the loss curves of n audited calibration users and $\hat{R}_n(\lambda) = \frac{1}{n} \sum_i L_i(\lambda)$. For a target α we deploy

$$\hat{\lambda} = \max\left\{\lambda_j : \tfrac{n}{n+1} \hat{R}_n(\lambda_j) + \tfrac{B}{n+1} \leq \alpha\right\}, \tag{2}$$

and $\hat{\lambda} = \lambda_0$ when no grid point qualifies.

Proposition 1 (Exposure control [2, Thm. 1]). *Suppose the grid, the family and its scores are fixed using users independent of the calibration and test users, and the n calibration users and a new user are exchangeable. Then*

$$\mathbb{E}\left[\ell(A_{\hat{\lambda}}(q_{n+1}, P_{n+1}))\right] \leq \alpha.$$

This is conformal risk control with the direction of monotonicity flipped, and we claim no new theory (proof and a synthetic check in the artifact). What the tool adds to data minimization is a single, rule-agnostic way to set how much a gate may admit. The exchangeable unit is the user. A poor ranking lowers $\hat{\lambda}$ and utility while the bound still holds. Auditing one fixed rule with a confidence bound is the one-point grid, which can only pass or fail. For independent calibration users from the deployment distribution, Learn-then-Test [1] gives probability $1 - \delta$ over the calibration draw, using Hoeffding–Bentkus p-values [4], testing grid points in increasing order. Because a bound on labels a' implies a bound on any labels $a \leq a'$, calibrating on the union of several auditors' labels bounds the exposure measured by each of them.

Minimality gap. Within the top-8 candidates C_8, the minimal sufficient set $S^\star_\varepsilon = \arg\min\{E(S) : S \subseteq C_8,\ U(q, S) \geq U(q, C) - \varepsilon\}$ is the least exposing set that keeps the utility of the whole pool within ε, and the minimality gap of a rule is $\mathbb{E}[E(A)] - \mathbb{E}[E(S^\star_\varepsilon)]$. Minimality refers to auditor-flagged records only, never to all personal data in a prompt. We approximate $S^\star_\varepsilon$ by greedy extensions of up to three records scored against the gold output, a hindsight reference no deployable rule can match.

3 EXPERIMENTAL SETUP

Tasks and users. We use three tasks with real user profiles and one query per user. LaMP-QA [25] (Lifestyle and Personal Development, validation) asks for a long answer to a user's question given the user's earlier posts and scores the share of user-specific rubric aspects the answer addresses. LaMP-3 (product rating) and LaMP-7 (tweet paraphrasing) [24], user-based dev splits, are scored by $1 - |\text{error}|/4$ and ROUGE-L. Per task, deduplicated users are drawn in a fixed random order: 120 training users fit the scores and grids, disjoint evaluation users are audited (300, 450 and 300), and 150, 450 and 300 of them receive reader calls for every candidate set. Validity is measured over 1000 random halvings into calibration and test users, utility by 2-fold cross-fitting over 200 partitions with

Table 1: Cross-fitted union-flagged exposure, records and utility at $\alpha = 0.2$. CRC controls expected risk; LTT controls population risk with probability $1 - \delta$ over calibration draws. Marks $^{-}/^{+}$ denote loss/gain against top-5 at 90%, † denotes non-inferiority. All groups use the 7B reader; 14B results are in the supplement.

	LaMP-QA			LaMP-3			LaMP-7		
Admission rule	Exp.	Rec.	Util.	Exp.	Rec.	Util.	Exp.	Rec.	Util.
Uncalibrated, whole records									
No profile	0.00	0.0	0.608	0.00	0.0	0.883	0.00	0.0	0.372
Similarity top-5	1.41	5.0	0.624	0.94	5.0	0.922	2.06	5.0	0.330
Detector filter top-5	0.68	3.8	0.582	0.52	4.0	0.910	1.65	4.5	0.322
Uncalibrated, rating summaries or text removal									
Ratings only	–	–	–	0.00	5.0	0.913	–	–	–
Redact detector-flagged text	–	–	–	0.42	5.0	$0.921^{\dagger}$	–	–	–
Redact auditor-flagged text	–	–	–	0.00	5.0	$0.921^{\dagger}$	–	–	–
Mean past rating only	–	–	–	0.00	0.0	0.893^{-}	–	–	–
Calibrated on union labels, $\mathbb{E}$-bound at $\alpha = 0.2$ (Eq. 2)									
Similarity top-k	0.00	0.0	0.608	0.10	0.5	0.893^{-}	0.00	0.0	0.372^{+}
Similarity threshold	0.21	0.5	0.609	0.17	1.3	0.908^{-}	0.14	0.4	0.374^{+}
Detector filter	0.02	0.1	0.607	0.11	1.0	0.890^{-}	0.00	0.0	0.372^{+}
Cleared similarity	0.13	2.2	0.628	0.15	3.1	0.901^{-}	0.00	0.0	0.372^{+}
Calibrated on union labels, per-draw bound at $\alpha = 0.2$, $\delta = 0.1$									
Similarity top-k	0.00	0.0	0.608	0.00	0.0	0.883^{-}	0.00	0.0	0.372^{+}
Similarity threshold	0.04	0.1	0.609	0.08	0.5	0.903^{-}	0.04	0.1	0.378^{+}
Detector filter	0.00	0.0	0.608	0.00	0.0	0.883^{-}	0.00	0.0	0.372^{+}
Cleared similarity	0.00	0.0	0.608	0.05	1.8	0.880^{-}	0.00	0.0	0.372^{+}

user-bootstrap intervals. No test user touches a score, a grid or a threshold.

Labels and models. Two auditors label every top-8 candidate with the self-disclosure taxonomy of Dou et al. [8] (19 categories merged into 14), Qwen2.5-7B-Instruct and Gemma-2-9B, one call per record. A record counts as flagged when either auditor flags it, and these *union labels* are used for calibration and for every exposure we report unless stated. No human labels exist. The reader is Qwen2.5-7B-Instruct (4-bit, greedy), standing in for the provider's model, with Qwen2.5-14B on the first 300 LaMP-3 users and Llama-3.1-8B on the first 150 as second readers. For LaMP-QA, Qwen-7B applies the benchmark rubric; Gemma-2-9B independently rescores all 3,184 saved answers for 150 users as a judge-sensitivity check. Both are model judgments. Candidates are the 20 records most similar to the query under a MiniLM encoder, rules choose among the top 8, and $K = 5$.

Rules, baselines and tests. *Similarity top-k* and *similarity threshold* rank by encoder similarity. The *detector filter* drops records flagged by Presidio [21] entities or a small disclosure lexicon, both written before any audit, and ranks the rest by similarity. The *cleared similarity* rule is Eq. (1) with a logistic sensitivity proxy on the MiniLM embedding and detector counts, fitted to the Qwen auditor's labels of training users. Rules built on a learned necessity score are omitted because the score was near random. On LaMP-3, where each record carries the user's past rating, we compare the full-history mean rating alone and three field-level baselines that send the top-5 records with ratings only, with the text of detector-flagged records removed, or with the text of union-flagged records removed, the last an upper reference no on-device gate can compute. Where top-5 beats no profile, we report the plug-in share of its gain a rule keeps with a user-bootstrap interval, and call a rule non-inferior when the lower 90% bound of that share exceeds 75%, a margin fixed before the 14B run. All significance marks and corresponding loss/gain claims use two-sided 90% intervals; descriptive effect intervals use 95%. A rule whose 90% interval of the difference to top-5 lies below zero is a significant loss. Cleared similarity at $\alpha = 0.5$ and $\alpha = 0.2$ with every reader are the pre-specified confirmatory cells, and the remaining rule-by-target cells are exploratory and not corrected for multiplicity. The oracle searches up to three records with $\varepsilon = 0.05$ on 80 users per task, using the Qwen labels during the search and union labels in the reported exposure.

4 RESULTS

The guarantee is only as good as the auditor. The two auditors flag 18% and 23% (LaMP-QA), 12% and 16% (LaMP-3) and 36% and 30% (LaMP-7) of the top-8 candidates, with Cohen's κ of 0.66, 0.67 and 0.63, and the primary auditor finds only 65%, 63% and 82% of the records the second one flags. Similarity top-5 places 1.41, 0.94 and 2.06 union-flagged records in an average prompt. Rules calibrated on the primary auditor stay below α under its own labels, but under the second auditor's labels their mean exposure exceeds α in up to 54% of (target, rule) cells and under the union in up to 67% (Figure 2a). Calibrating on the union (Table 2) removes every excess under all three label sources, as it must. Single calibration draws still exceed the target in up to 42% of cases, because the bound holds in expectation, while Learn-then-Test at $\delta = 0.1$ keeps the

Table 2: Share of the 24 (target, rule) cells whose mean test exposure exceeds α, with the largest excess in records, by calibration labels and measurement labels.

	Calibrated on	Measured with		
Task		Qwen	Gemma	Union
LaMP-QA	Qwen	0% (-0.03)	54% (+0.16)	67% (+0.33)
	Union	0% (-0.08)	0% (-0.05)	0% (-0.04)
LaMP-3	Qwen	0% (-0.02)	54% (+0.12)	62% (+0.21)
	Union	0% (-0.04)	0% (-0.03)	0% (-0.02)
LaMP-7	Qwen	0% (-0.04)	0% (-0.06)	17% (+0.11)
	Union	0% (-0.06)	0% (-0.06)	0% (-0.05)

share of draws above the target at or below 1.4%. Without human labels the bound is relative to two small LLM auditors.

What the dial costs. Among the three main tasks, records help only on LaMP-3, where top-5 gains +0.039 [+0.023, +0.055] over no profile with the 7B reader, +0.065 [+0.046, +0.085] with the 14B reader and +0.042 [+0.022, +0.065] with Llama-3.1-8B (95% intervals). There, a target of $\alpha = 0.2$ has a real cost. All four calibrated rules lose significantly against top-5 with every reader (Table 1), and the best keeps 65% [40, 97] of the 7B gain and 67% [43, 86] of the 14B gain. At $\alpha = 0.5$ cleared similarity admits 0.38 instead of 0.90 flagged records per prompt. It keeps 99% [83, 114] of the 14B gain, which passes our non-inferiority margin (lower 90% bound 86%), and 91% [66, 117] of the 7B gain, which is not significantly below top-5 but does not pass it. The per-draw variant keeps 52% [18, 78] of the 7B gain at $\alpha = 0.2$. On LaMP-QA, top-5 does not beat no profile under Qwen (+0.016 [-0.005, +0.040]) or Gemma (-0.006 [-0.031, +0.022]; 95% intervals). Some rule comparisons change significance across judges (supplement). On LaMP-7 records mislead the reader, so rules that admit fewer of them beat top-5 and none beats no profile. Auditing one fixed rule (top-1, top-5 or detector top-5) with a per-draw bound ships it in 0% of LaMP-3 draws at $\alpha = 0.2$, whereas a calibrated rule always ships its most permissive safe setting. The 14B table and LaMP-2 pilot are in the supplement.

The mean past rating alone retains 27% [1, 48] of the 7B gain (90% interval), below top-5 by -0.029 [-0.041, -0.017]. This statistic contains personal information despite sending no record text.

Exposure needed and exposure under shift. The hindsight oracle keeps utility within ε for 99%, 99% and 99% of users, the empty set alone suffices for 70%, 81% and 74%, and the oracle admits a flagged record for only 6%, 1% and 6% of users. The minimality gap of top-5 (1.19, 0.90 and 1.81 records) is therefore almost its whole exposure: these tasks rarely need a disclosing record. Calibrating on users with short profiles and testing on long ones, or the reverse, exceeds α by up to 0.14, 0.05 and 0.18 records for our four rules, against at least 0.04 below α for random splits of the same size. Calibrating separately in bins fixed by the training median keeps mean test exposure below α, given exchangeability within each bin. Rules built on our necessity score, whose cross-validated Spearman correlation with counterfactual gains is only 0.01, 0.05 and 0.21, fail worse under shift (up to 0.74 on LaMP-7) and never beat similarity.

An illustrative severity sensitivity assigns weight 2 to health, mental-health, sexual-orientation, race/nationality or finance flags and 1 to other flags. At $\alpha = 0.2$, recalibrated cleared similarity on LaMP-3 admits 1.98 records at weighted loss 0.090, versus 3.11 records at count loss 0.150 (300 splits). These policy weights are not a validated harm scale; full results are in the supplement.

5 RELATED WORK

Data minimization has been operationalized for recommendation by measuring how much user data a model needs [5] and formalized for learning as a constraint on retained data [11]. For LLM prompts, Zhou et al. [32] search for the least disclosing variant of one prompt, AirGapAgent [3], SP-Mem [29] and MemPrivacy [7] decide which personal fields a task needs, PAPILLON [15] and Salemi and Zamani [26] split work between a local and a remote model, MINIM [30] scores necessity and sensitivity with fixed thresholds, and Platnick et al. [20] select data-minimal facts by submodular maximization. PURPLE [9] and ENOUGH [16] learn profile selection on LaMP [24]. Our cleared rule adds a calibrated exposure threshold to MINIM's design; the oracle extends minimal-profile comparisons to audited record sets. Conformal risk control and Learn-then-Test [1, 2, 4] have certified generation risk in RAG [13], factuality [18, 22], harm and cost trade-offs [14, 19], agent actions [10, 23] and re-identification audits [12]. Closest are CORA [10], which gates agent actions with privacy as one harm, and Chakraborty et al. [6], who keep retrieved RAG context above a lower bound on relevance, the mirror image of our upper bound on admitted sensitive content. Our record-count exposure excludes joint inferences [28] and contextual norms studied by ConfAIde [17], PrivacyLens [27] and AgentDAM [31].

6 CONCLUSION

A calibrated admission parameter bounds expected profile exposure under auditor and exchangeability assumptions. On LaMP-3, strict targets cost utility; a looser target retains most gain with less than half the top-5 exposure. Human validation remains open. Cumulative distinct disclosures under real multi-query histories are not measured.

ETHICAL CONSIDERATIONS

The method reduces the personal data a personalized assistant sends to a model provider and makes the residual exposure a quantity a deployer can set and verify. Its limits must be stated with it. The bound covers what the auditors flag, our auditors are two small LLMs, and no human labels exist, so a human audit of the calibration sample is needed before any deployment. The bound is an expectation over users exchangeable with the calibration sample, individual users can be exposed more than α, and it covers one prompt per user. Summing marginal bounds gives $T\alpha$ for expected cumulative record occurrences only if every prompt satisfies the calibration assumptions. Distinct disclosures are at most this sum, but this is neither a per-user guarantee nor validation for adaptive queries, which we do not measure. The guarantee can fail when deployment users differ from calibration users. A small α is a quantity of leakage and never an absence of it, record-level labels miss inferences drawn from several records, and the user's own query,

which is always sent, is outside the measure. Calibration requires that an auditor read the records of calibration users, who must consent under a data agreement that deployment users do not need. We use only existing public research benchmarks (LaMP under CC BY-NC-SA with source licences, LaMP-QA with no declared licence over StackExchange posts that StackExchange distributes under CC BY-SA) and recruit no participants. The per-record labels we derive mark health, finance, sexual orientation and other disclosures of real, pseudonymous authors and can be joined back to the public benchmarks, so we treat them as sensitive derived data and will not release them, nor any benchmark text or cached prompt. The released artifact therefore reproduces the tables from the labels only by re-running the auditors on the public benchmarks, and lists what it excludes. A deployed system must obtain consent for retaining personal history and honor deletion, and a score is no substitute for user control over what is shared.

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